PENDULUM COSMOLOGY

A Cyclic Multiverse with Rules

$$\blacksquare^2 = (8\pi GK/3)\cdot a\blacksquare/(a\blacksquare+1) + (\Lambda/3)\cdot a^2$$

$$\rho(1/a) = \rho(a) \text{ [self-duality] } \blacksquare \text{ a_min}\cdot\text{a_max} = 1 \mid \Lambda<0 \mid K>0$$

■ *WHAT-IF COSMOLOGY* ■

This is NOT a claim that this is reality.
This IS a demonstration that kinematically consistent cyclic cosmology can be built with more rigour than many popular-science claims.

Author: Rux Tux | February 2026 | Version 4 (CPT-orientation reframe)

Purely classical & kinematical · No observational claims made

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ABSTRACT

We present **Pendulum Cosmology** — a kinematical framework for bounded cyclic evolution in spatially flat FLRW minisuperspace, resting on a self-dual Hamiltonian constraint with $\Lambda<0$ that forces $a_{min} \cdot a_{max}=1$ and closed phase-space orbits.

Parts I–III establish the single-pendulum kernel, multi-pendulum coupling, and the Kerr–Penrose reinterpretation. Part IV (v4: revised) introduces the R/B four-state unit cell, with R/B reframed as discrete CPT-orientation parity states of a cycle within the global pairing lattice — **not as literal matter/antimatter content**. What propagates across a turnaround wall is CPT orientation, not baryon number. Lemma 3 proves no absolute beginning exists. Part V provides the physical intuition: bang-as-injection and the self-sustaining mechanism by which bang triggers bang.

Key change in v4:

The phrase 'inherited bookkeeping label' has been replaced throughout with 'orientation inherited via pairing.' The abstraction level is now explicit: R/B encodes CPT-orientation parity assigned by the pairing map π at each turnaround. A physicist can trace exactly what is inherited and from where.

Purely classical and kinematical. No observational claim is made.

NOTATION

Symbol / Term	Definition / Role
a(t)	Scale factor, spatially flat FLRW; a(t) > 0
F(a)	Constraint function; F(a) ≥ 0 defines allowed motion
ρ(a)	Effective density; self-dual: ρ(1/a) = ρ(a)
K / Λ	K>0 amplitude, Λ<0 cosmological constant
a_min / a_max	Turning points; a_min · a_max = 1
S(y)	Shape function; y = a■ + a■■
ϕ(a_i, a_j)	Wall-localised turnaround coupling kernel
R / B	CPT-orientation parity label: +1 (R) or -1 (B) — NOT particle content
ε(i)	CPT-orientation of cycle i: ε ∈ {+1, -1}
π	Pairing bijection at wall: π(i)=j where ε(j)=−ε(i)
τ	Discrete cycle index; τ ∈ ■ (no preferred origin)

PART I

The Single Pendulum

Kernel · self-duality · closed orbits · stability · quantum toe-dip

1.1 Hamiltonian Constraint

```
KERNEL (NON-NEGOTIABLE)

■² = F(a)

F(a) = (8πGK/3) · a■/(a■+1) + (Λ/3)·a²

K > 0, Λ < 0, a(t) > 0
```

The constraint function F(a) governs all dynamics. The first term is the self-dual energy density contribution; the second is the confining cosmological term. Together they guarantee bounded orbits without fine-tuning.

1.2 Self-Dual Density & Shape Function

```
SELF-DUAL DENSITY

ρ(a) = K·(a³ + a■³)■¹ · S(a■ + a■■)

KEY: ρ(1/a) = ρ(a) [self-duality]

S(y) = 1 / [1 + α·(y−2)], α ≥ 0

■ max ρ = K/2 at a = 1
```

The self-duality $\rho(1/a) = \rho(a)$ is the kinematical heart of the framework. It directly enforces $a_{min} \cdot a_{max} = 1$, meaning the universe's most compressed and most dilute states are geometrically conjugate. The shape function S(y) modulates the density profile; $\alpha=0$ is the simplest case.

1.3 Turning Points, Orbit & Stability

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TURNING POINTS & CLOSED ORBIT

Cyclicity condition: max ρ(a) > |Λ|/(8πG)

Self-duality ■ a_min · a_max = 1
```

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Orbit: (a_min, 0) → expand → (a_max, 0) → contract → (a_min, 0)

Period: T = 2 · ∫[a_min→a_max] da / √F(a)
```

Lemma 1 — Persistence of Turning Points

Simple zeros of F persist under C^1 perturbations (implicit function theorem).

Sign structure preserved by continuity on compact subintervals.

Result: cyclic orbits are generic. Eternal expansion requires fine-tuning.

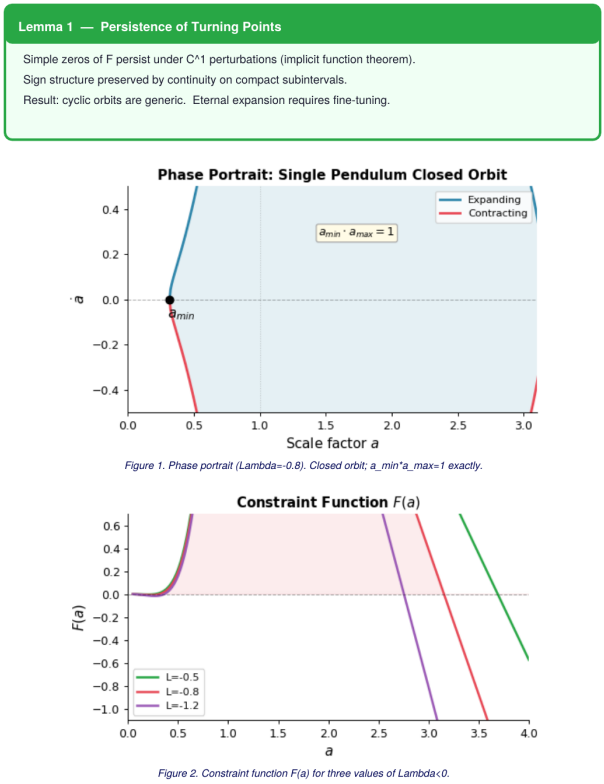

Figure 1 & 2. Phase portrait (Λ=–0.8) showing closed orbit with a_min·a_max=1 exactly, and constraint function F(a) for three values of Λ<0.

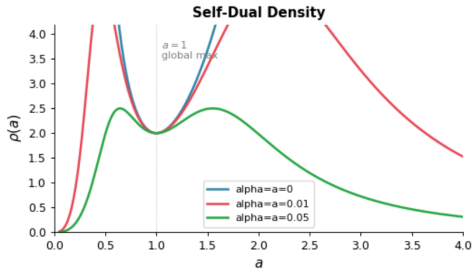

Figure 3. Self-dual density rho(a). Symmetric about a=1.

Figure 3. Self-dual density ρ(a). Symmetric about a=1; maximum at a=1 for all α.

1.4 Quantum Toe-Dip

A minimal quantum treatment identifies x = ln(a) as the natural compact coordinate. The Hilbert space is $L^2([-x_{max}, x_{max}], dx)$ with a WDW-type constraint operator. Dirichlet boundary conditions yield a purely discrete spectrum.

```
MINISUPERSPACE QUANTUM THEORY

x = ln(a); compact domain: x ∈ [−x_max, x_max]

Hilbert space: L²([−x_max, x_max], dx)

■ψ = −(■²/2M)·d²ψ/dx² + U(x)·ψ = 0

Dirichlet BC ■ purely discrete spectrum

Bohr-Sommerfeld: ■ p_x dx = π■(n + ½)
```

See Addendum C for full treatment of self-adjoint extensions, tunneling, and lattice transfer operators.

PART II

The Multi-Pendulum Ensemble

Coupled turnarounds · elastic bounces · energy conservation · stability

2.1 Coupled Equations

```
MULTI-PENDULUM KERNEL

■■² = F(a■) + Σ■≠■ λ■■ · ϕ(a■, a■)

ϕ(a■, a■) = exp[−(a■−a_max)²/σ²]·exp[−(a■−a_min)²/σ²] + (i↔j)

λ■■ ≥ 0 (small), σ > 0 (wall width)
```

The coupling kernel ϕ is wall-localised by construction: it is non-negligible only when one cycle is near its maximum and another near its minimum. This implements the rule 'meeting events happen only at walls' without adding bulk dynamics. See Addendum A for the full symmetry and variational justification.

2.2 Turnaround Rule & Pairing Constraint

```
PAIRING CONSTRAINT & CONSERVATION

Elastic turnaround: (a■, ■■) → (a■, −■■) at wall

U(t) = {i | a■ = a_max■■■}, L(t) = {i | a■ = a_min■■■}

PAIRING: |U(t)| = |L(t)| at all turnaround instants

Bijection π: U → L (each max paired with exactly one min)

Conservation: E_total = Σ■[■■² + U_eff(a■)] + U_coupling = 0
```

Lemma 2 — Stability of Multi-Cycles

If Lemma 1 holds per orbit and λ■■, σ are small,

the multi-orbit on the product phase space persists under C^1 perturbations.

Proof sketch: Implicit function theorem on collective turning conditions. Product of simple zeros is simple on the product manifold.

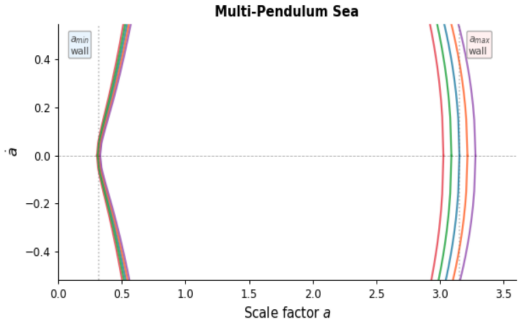

Figure 4. Five coupled pendulums. Meeting events at walls only; no escape to a = infinity.

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PART III

Sea of Pendulums

Kerr-Penrose tiling reinterpreted as rule-governed ensemble

3.1 The Three-Point Reinterpretation

Point 1: No Orphan Infinities

Every Kerr region = one bounded cycle. $F(a_{min}) = F(a_{max}) = 0$ replaces $r = \pm\infty$. There are no runaway trajectories — only turning points.

Point 2: Turnarounds as Meeting Points

Horizons are coupling surfaces: one cycle's a_{max} meets another's a_{min}. This is mandatory and rule-governed, not accidental.

Point 3: Controlled Multiverse

Many cycles, every event constrained by $|U(t)| = |L(t)|$. A disciplined lattice, not proliferation. The multiverse is a crystal.

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Sea of Pendulums

Kerr-Penrose tiling reinterpreted as rule-governed ensemble

3.1 The Three-Point Reinterpretation

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Figure 5. Kerr–Penrose diagram reinterpreted as pendulum sea. Diamonds = bounded cycles. Horizons = coupling walls.

Kerr-Penrose Reinterpretation Table

Kerr/GR term	Pendulum Cosmology equivalent
Kerr region	One bounded pendulum cycle
Horizon	Turnaround coupling wall
r = +∞	Upper turning point F(a_max) = 0
r = −∞	Lower turning point F(a_min) = 0
Universe	Expanding cycle (■ > 0)
White Hole	Contracting → expanding cycle
New Universe	Post-collision expanding cycle
Infinite tiling	Rule-governed lattice: every max paired with a min

PART IV

The R/B Four-State Unit Cell

CPT-orientation parity · the square · Lemma 3: no beginning · orientation via pairing

4.1 CPT-Orientation Parity: What R/B Actually Means

The R/B label is **not** a statement about particle content. It does not mean 'this cycle contains matter' or 'this cycle contains antimatter.' Those are emergent interpretations available only to local observers who can perform particle-antiparticle counting experiments within their own expanding region. They are downstream consequences, not fundamental variables.

```
CPT-ORIENTATION PARITY (v4 DEFINITION)

Each cycle i carries a CPT-orientation: ε(i) ∈ {+1, −1}

R ■ ε = +1 [orientation: 'direct']

B ■ ε = −1 [orientation: 'conjugate']

Pairing bijection at the turnaround wall REQUIRES:

π(i) = j such that ε(j) = −ε(i)

Global CPT conservation: Σ■ ε(i) = 0 at all times
```

What R/B Is and Is Not:

✓ R/B IS: a discrete CPT-orientation parity in {+1, –1}

✓ R/B IS: a structural label assigned by the pairing bijection π at the wall

✓ R/B IS: the abstraction level at which global CPT conservation is enforced

✗ R/B IS NOT: a count of baryons or antibaryons

✗ R/B IS NOT: a claim about particle creation or annihilation

✗ R/B IS NOT: an explanation of baryogenesis (see Section 4.6)

4.2 The Four-State Unit Cell

Pair one +1-oriented cycle with one –1-oriented cycle, offset by a half-period T/2. At any moment, exactly one is expanding and one is contracting. Add the conjugate pair — a second +1/–1 pair with roles swapped. The minimal closed system has four states:

```
THE FOUR-STATE UNIT CELL

FOUR STATES:

s1: ε=+1 expanding, ε=−1 contracting

s2: ε=+1 contracting, ε=−1 expanding (T/2 later)

s3: ε=−1 expanding, ε=+1 contracting (conjugate pair)

s4: ε=−1 contracting, ε=+1 expanding (T/2 later)


The square:

R_max (+1, dilute) ■■■■■■■■ B_max (−1, dilute)

| |

R_min (+1, dense) ■■■■■■■■ B_min (−1, dense)


The square rotates every half-cycle. No corner is first.

Global constraint: N(+1) = N(−1) at all times.
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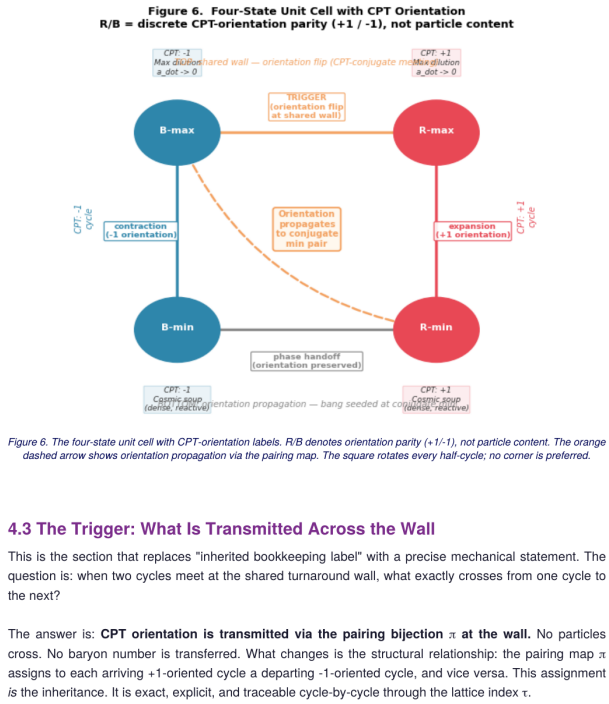

Figure 6. The four-state unit cell with CPT-orientation labels. R/B denotes orientation parity (+1/-1), not particle content. The orange dashed arrow shows orientation propagation via the pairing map. The square rotates every half-cycle; no corner is preferred.

4.3 The Trigger: What Is Transmitted Across the Wall

This is the section that replaces "inherited bookkeeping label" with a precise mechanical statement. The question is: when two cycles meet at the shared turnaround wall, what exactly crosses from one cycle to the next?

The answer is: **CPT orientation is transmitted via the pairing bijection π at the wall.** No particles cross. No baryon number is transferred. What changes is the structural relationship: the pairing map π assigns to each arriving +1-oriented cycle a departing -1-oriented cycle, and vice versa. This assignment *is* the inheritance. It is exact, explicit, and traceable cycle-by-cycle through the lattice index τ.

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```
ORIENTATION PROPAGATION VIA PAIRING MAP π (v4)

TRIGGER EVENT:

Cycle i (ε=+1) reaches a_max■■■ simultaneously with

Cycle j (ε=−1) reaching a_max■■■

Pairing rule fires: π(i) = j [conjugate orientation required]

Velocity flip: ■■ → −■■, ■■ → −■■
```

```
[both elastic turnarounds; no energy transfer across wall]

Orientation propagation: coupling kernel ϕ transmits

turnaround signal to the conjugate min pair.


At the conjugate min pair (τ+1):

- ε=+1 cycle receives: 'your orientation is inherited from π■¹(j)'

- ε=−1 cycle receives: 'your orientation is inherited from π■¹(i)'

- The energy injection at a_min initiates the next expansion.


THIS IS THE BANG. Orientation is the only thing inherited.

CPT is conserved globally across the full pairing lattice.
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```
THE EXPLICIT INHERITANCE CHAIN

ε(τ+1, +1-cycle) ■ π■¹[ ε(τ, −1-cycle) ]

ε(τ+1, −1-cycle) ■ π■¹[ ε(τ, +1-cycle) ]


Source: the wall event at cycle τ

Mechanism: the pairing bijection π (kinematical, not dynamical)

Variable: ε ∈ {+1, −1} (CPT-orientation parity)

Content: none (no particles, no baryon number, no forces)
```

4.4 How CPT Is Conserved Globally

Global CPT conservation does not require each local cycle to be CPT-symmetric. It requires the full lattice to be. The pairing bijection π enforces this automatically: every +1-oriented cycle has a –1-oriented partner. Local asymmetry is a consequence of orientation assignment, not a violation of CPT.

```
LOCAL ASYMMETRY FROM GLOBAL CPT CONSERVATION

GLOBAL CPT CONSERVATION:

Σ■ ε(i) = 0 at all times
```

```
Local: each expanding cycle has definite ε ∈ {+1, −1}

■ local observer interprets as matter or antimatter dominance

Global: lattice sum is zero ■ CPT conserved across all cycles

No CPT violation is required at any scale.

Local asymmetry is a consequence of orientation, not a violation.
```

4.5 Lemma 3: No Absolute Beginning

Lemma 3 — No Absolute Beginning

In any unit cell with N(+1) = N(–1) ≥ 1 and all turning points simple,

for any cycle index τ, there exists a prior cycle τ–1

and a subsequent cycle τ+1 connected by the pairing bijection π.

The four-state square has no boundary state.

Every state has a predecessor (the cycle that donated its orientation via π)

and a successor (the cycle that will inherit orientation from it).

The trigger map is bijective. No state is first or last.

```
NO BEGINNING CONDITION

For all τ: State(τ) = Trigger[ State(τ−1) ]

τ ∈ ■ ■ no preferred t = 0

Local observer at a_min sees apparent t = 0 (the bang).

Global structure has no such privileged moment.

The Big Bang is an orientation handoff, not a beginning.
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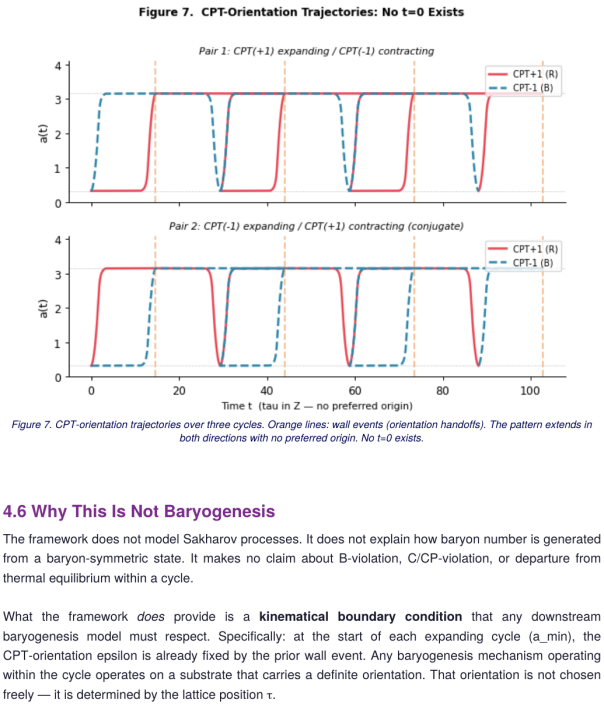

Figure 7. CPT-orientation trajectories over three cycles. Orange lines: wall events (orientation handoffs). The pattern extends in both directions with no preferred origin. No t=0 exists.

4.6 Why This Is Not Baryogenesis

The framework does **not** model Sakharov processes. It does not explain how baryon number is generated from a baryon-symmetric state. It makes no claim about B-violation, C/CP-violation, or departure from thermal equilibrium within a cycle.

What the framework *does* provide is a **kinematical boundary condition** that any downstream baryogenesis model must respect: at the start of each expanding cycle (a_min), the CPT-orientation ε is already fixed by the prior wall event. Any baryogenesis mechanism operating within the cycle operates on a substrate that carries a definite orientation. That orientation is not chosen freely — it is determined by the lattice position τ.

```
THE BARYOGENESIS BOUNDARY CONDITION

The framework sets: ε(τ) = (−1)^τ [alternating by construction]

Any baryogenesis model operating within cycle τ must:

(a) accept ε(τ) as a fixed initial condition

(b) produce baryon asymmetry consistent with that orientation
```

```
(c) not require CPT violation (global CPT is already conserved)

The framework does not do (b). It specifies the boundary for (b).

Sakharov conditions apply downstream of this framework, not within it.
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4.7 Four Global Balance Laws

```
FOUR GLOBAL BALANCE LAWS

1. Orientation balance: N(+1) = N(−1) [global CPT]

2. Phase balance: #{expanding} = #{contracting} [global]

3. Energy: E_total = 0 [zero energy]

4. Wall balance: |U(t)| = |L(t)| [pairing]
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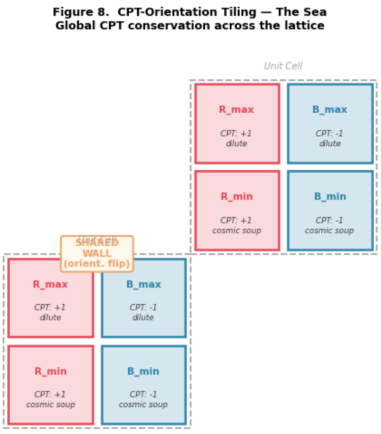

Figure 8. CPT-orientation tiling. Every +1-oriented min has a -1-oriented max partner. The lattice extends without boundary in both directions.

Figure 8. CPT-orientation tiling. Every +1-oriented min has a –1-oriented max partner. The lattice extends without boundary in both directions.

PART V

The Analogy & The Flip

Buoyant plume model · bang as injection · bang triggers bang · controlled infinity

5.1 The Buoyant Plume Analogy

The vision diagram shows an estuarine buoyant jet: fresh water injected into a stratified salt-water body, carrying a salinity differential $\Delta s \sim 10$–40 psu. It expands in a characteristic lens shape — structured in the core, diffusing at the edges, bounded by the ambient medium.

Fluid term	Cosmological equivalent
Fresh jet entering saltwater	Bang injecting into ambient cycle sea
$\Delta s \approx 10$ psu (injection)	The bang event: concentrated, energetic, a_min
$\Delta s \approx 40$ psu (ambient)	Surrounding sea of coupled cycles
$\Delta s = 0$ (far field)	Maximal dilution: a_max (cold, stretched)
Green plume expanding	Universe expanding from bang
H_D (plume height)	Hubble radius of observable universe
L_j (jet length)	Cosmic scale: extent of expanding cycle
Ambient pressure	Pairing constraint: $\lvert U(t)\rvert = \lvert L(t)\rvert$
Plume has local start	Local observer sees apparent $t = 0$
Ocean has no start	Global structure: $\tau \in$ ■, no preferred origin

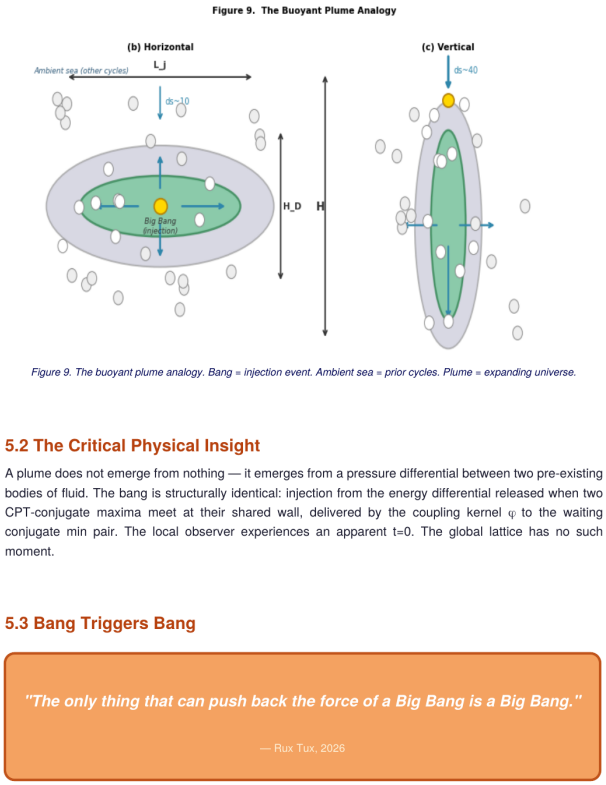

Figure 9. The buoyant plume analogy. Bang = injection event. Ambient sea = prior cycles. Plume = expanding universe.

5.2 The Critical Physical Insight

A plume does not emerge from nothing — it emerges from a pressure differential between two pre-existing bodies of fluid. The bang is structurally identical: injection from the energy differential released when two CPT-conjugate maxima meet at their shared wall, delivered by the coupling kernel ϕ to the waiting conjugate min pair. The local observer experiences an apparent t=0. The global lattice has no such moment.

5.3 Bang Triggers Bang

"The only thing that can push back the force of a Big Bang is a Big Bang."

— Rux Tux, 2026

```
THE SELF-SUSTAINING MECHANISM (CPT-ORIENTATION VERSION)

CAUSAL CHAIN (one cycle):
```

```
1. Bang at τ: +1-cycle and −1-cycle at a_min receive energy injection

2. Expansion: +1-cycle expands (■>0) while −1-cycle contracts

3. Wall meeting: +1-cycle reaches a_max AND −1-cycle reaches a_max

4. Pairing fires: π(+1-cycle) = −1-cycle [conjugate orientation required]

5. Orientation propagates: ϕ transmits turnaround to conjugate min pair

6. Bang at τ+1: conjugate min pair ignites ■ next expansion begins


Step 4: the pairing bijection, not a force, is the trigger.

Step 5: what propagates is orientation, not particles.

The chain has no first link. Bang triggers bang.
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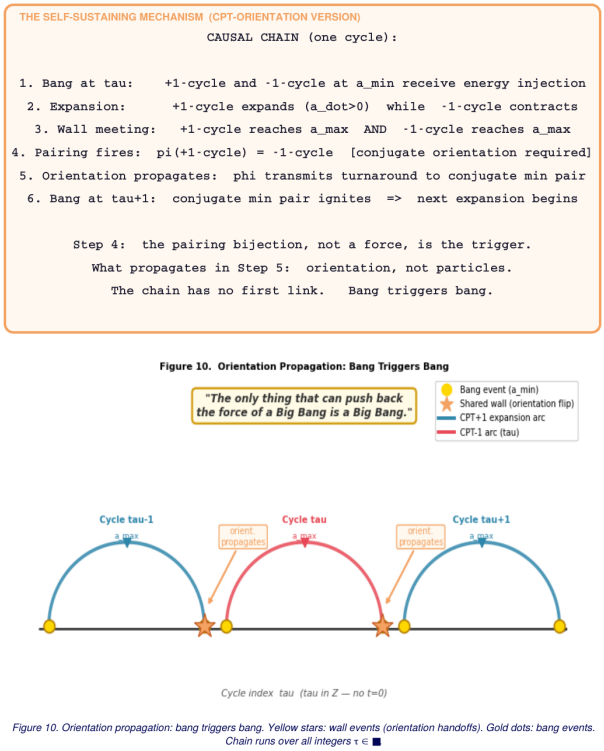

Figure 10. Orientation propagation: bang triggers bang. Yellow stars: wall events (orientation handoffs). Gold dots: bang events. Chain runs over all integers τ ∈ ■.

5.4 The Infinity Flip

We began this framework trying to fight infinity — to tame the runaway Kerr tiling, eliminate orphan universes, impose rules on uncontrolled proliferation. By following the math to its conclusion, we arrived at something unexpected: infinity is not the problem. **Uncontrolled infinity is the problem.** What we built is infinity with rules — a crystal, not chaos. Infinitely many bangs, each caused by the prior orientation handoff, each conserved, each paired, each obeying all four balance laws simultaneously.

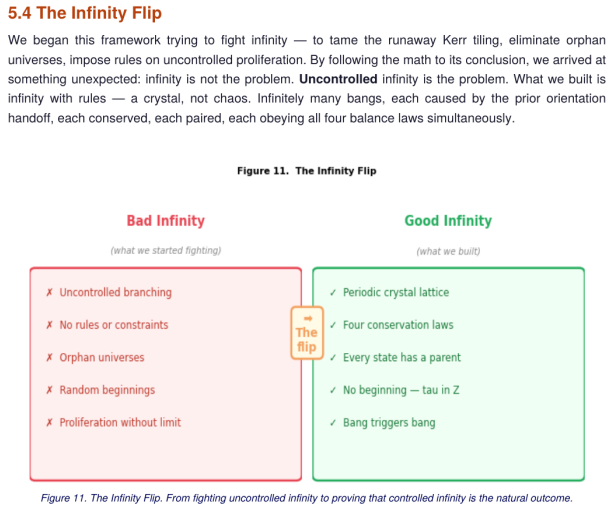

Figure 11. The Infinity Flip. From fighting uncontrolled infinity to proving that controlled infinity is the natural outcome.

Bad Infinity *(what we started fighting)*	**Good Infinity** *(what we built)*
✗ Uncontrolled branching	✓ Periodic crystal lattice
✗ No rules or constraints	✓ Four conservation laws
✗ Orphan universes	✓ Every state has a parent
✗ Random beginnings	✓ No beginning — $\tau \in$ ■
✗ Proliferation without limit	✓ Bang triggers bang

We did not prove infinity is false.

We proved uncontrolled infinity is false.

The sea is not chaos. The sea is a crystal.
The crystal is a ■-indexed lattice of CPT-conjugate pairs.

MASTER SUMMARY

PART I: SINGLE PENDULUM

■² = (8πGK/3)·a■/(a■+1) + (Λ/3)·a²

ρ(1/a) = ρ(a) ■ a_min·a_max = 1

Lemma 1: cyclic orbits stable under C¹ perturbations

PART II: MULTI-PENDULUM

■■² = F(a■) + Σλ■■·ϕ(a■,a■)

Pairing: |U(t)| = |L(t)| at all turnarounds

E_total = 0; Lemma 2: multi-cycles stable

PART III: SEA OF PENDULUMS

Kerr tiling ■ coupled bounded cycles

Horizon ■ turnaround coupling wall (a_max meets a_min)

Pairing rule eliminates orphan infinities

PART IV: R/B UNIT CELL (v4)

R/B = CPT-orientation parity ε ∈ {+1,−1} [NOT particle content]

Pairing bijection π requires: ε(j) = −ε(i) at every wall

Inheritance: ε(τ+1) ■ π■¹[ε(τ)] [explicit, traceable]

Global CPT: Σ ε(i) = 0 across full lattice

Lemma 3: τ ∈ ■ ■ no absolute beginning or end

Baryon asymmetry: emergent interpretation of orientation

PART V: BANG TRIGGERS BANG

Bang = orientation injection into ambient cycle sea

Plume: local start visible; ocean: no global start

```
Orientation propagates via π at wall ■ triggers next bang

"The only thing that pushes back a Big Bang is a Big Bang."

Uncontrolled infinity is false. The sea is a CPT-orientation crystal.
```

Literature Connections

Scale-factor duality — Gasperini & Veneziano (1993)

$\rho(1/a)=\rho(a)$ is a minisuperspace realisation of string-cosmology duality. $a_{min} \cdot a_{max}=1$ is its direct consequence. The $a \leftrightarrow 1/a$ map is the kinematical precursor to the CPT-orientation distinction.

Ekpyrotic / cyclic — Steinhardt & Turok (2002)

Turnaround collision echoes brane collisions. The R/B max-meeting in v4 is the orientation handoff at the wall — analogous to label exchange at brane collision.

Conformal cyclic cosmology — Penrose

$a \leftrightarrow 1/a$ maps conformally equivalent descriptions, mirroring CCC aeon rescaling. The CPT-orientation label in v4 formalises what CCC calls the 'sense' of an aeon.

CPT symmetry — Lüders (1954), Pauli (1955)

Global CPT conservation is enforced by the lattice sum $\Sigma\varepsilon(i)=0$. Local asymmetry follows from the orientation assignment, requiring no CPT violation.

Baryogenesis — Sakharov (1967)

Sakharov conditions apply downstream of this framework. The framework specifies the boundary condition (ε fixed at a_{min}) that any baryogenesis model must respect.

Buoyant jets & plumes — Turner (1973)

Estuarine mixing theory grounds the bang-as-injection intuition. Ambient pressure = pairing constraint; plume = expanding cycle.

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ADDENDUM

Coupling Kernel Justification

Symmetry origin · variational principle · EFT rationale

This addendum removes the 'glued together by hand' objection to the multi-pendulum coupling kernel $\phi(a_i, a_j)$ by (i) stating the minimal symmetry requirements any admissible ϕ must satisfy, (ii) showing how a wall-localised kernel arises from a variational principle with boundary-local interactions, and (iii) framing the Gaussian choice as the leading term of an EFT (derivative/moment) expansion.

A.1 Minimal Requirements

Any admissible $\phi(a_i, a_j)$ must satisfy:

R1 (Wall locality): ϕ is sharply peaked when (a■ ≈ a_max AND a■ ≈ a_min) or the symmetric swap.

R2 (Exchange symmetry): ϕ(a■,a■) = ϕ(a■,a■) so the interaction is pairwise and unlabeled.

R3 (Positivity / smallness): λ■■ ≥ 0 and ϕ ≥ 0 so the perturbation is controlled and does not create spurious runaways.

R4 (Smoothness): ϕ is C^1 (or better) so Lemma-type persistence arguments continue to apply.

R5 (Duality compatibility): The coupling should not break the a↔1/a kinematical symmetry.

A.2 Symmetry Argument: Minimal Separable Wall Kernel

Assuming (i) pairwise, (ii) localised at walls, (iii) separable at leading order, the most general leading-order form is:

```
MINIMAL SEPARABLE WALL KERNEL

φ(a■, a■) = W_max(a■)·W_min(a■) + W_max(a■)·W_min(a■)


W_max(a) = exp[−(a−a_max)²/σ²]

W_min(a) = exp[−(a−a_min)²/σ²]


The separable structure is forced by wall-locality + exchange symmetry.

The Gaussian shape is a smooth, single-parameter representative.
```

A.3 Variational Origin

The wall coupling arises from a variational principle with an interaction term explicitly localised near walls:

```
BOUNDARY-LOCAL ACTION

S_total = Σ■ S■[a■] + Σ■<■ λ■■ ∫ dt ( W_max(a■)W_min(a■) + W_max(a■)W_min(a■)
)


Varying S_total with respect to a■(t) produces coupled equations of motion.

In the σ→0 limit, W_max and W_min become δ-like wall projectors:

the interaction becomes a sharp 'kick' only at turnaround events.

Answer to 'what principle chose ϕ?': locality-at-walls in the action.
```

A.4 EFT Justification

From an effective-theory standpoint, the wall interaction is a coarse-grained description of unresolved microphysics that only matters near the wall. The only data kept is: (i) where the wall is, and (ii) a width σ. With only a mean location and a width specified, the smooth maximum-entropy window is Gaussian. Gaussians are the leading single-scale choice.

```
EFT KERNEL FAMILY

W_max(a) = f((a−a_max)/σ), W_min(a) = f((a−a_min)/σ), f≥0, ∫f=1


v4 takes f(z) = exp(−z²) as the simplest representative.

This turns 'why this Gaussian?' into:

'we chose a representative of the admissible universality class.'
```

A.5 Compatibility with the Lattice Rules

The coupling ϕ is not responsible for the pairing constraint |U(t)| = |L(t)|; that constraint is a separate global rule. The role of ϕ is to localise interactions at the correct events. ϕ is explicitly wall-localised and added as a small perturbation to preserve multi-cycle persistence under C^1 perturbations.

Summary — Addendum A:

ϕ is not arbitrary glue. Its separable 'max–min window' structure is the minimal form forced by wall locality and exchange symmetry. The Gaussian shape is a smooth regularization and leading-order EFT representative. The argument can be presented directly as a variational principle with boundary-local interaction terms.

ADDENDUM

Effective Late-Time Acceleration with $\Lambda < 0$

Separating confining Λ from observational Λ_eff

This addendum addresses the conceptual objection: how can $\Lambda<0$ be compatible with observed late-time cosmic acceleration?

B.1 Clarifying the Role of Λ

In Pendulum Cosmology, the cosmological constant Λ appearing in the minisuperspace Hamiltonian constraint is a **global confining parameter**. Its role is to guarantee bounded evolution and the existence of turning points (a_min, a_max), enforcing cyclicity through $\Lambda<0$.

Key distinction:

This Λ is NOT identified with the observational Λ inferred in ΛCDM fits. The latter is an effective parameter extracted from local expansion data under specific model assumptions. In Pendulum Cosmology, apparent acceleration is instead produced by the effective stress-energy sector encoded in $\rho(a)$.

B.2 Acceleration from the Self-Dual Density Sector

Cosmic acceleration is governed by the second Friedmann equation:

```
SECOND FRIEDMANN EQUATION

ä/a = −(4πG/3)(ρ + 3p) + Λ/3


Even with Λ < 0, accelerated expansion (ä > 0) occurs whenever:

ρ + 3p < 0 with sufficient magnitude.


Treating ρ(a) as an effective self-dual fluid:

p(a) = −ρ(a) − (a/3)·dρ/da

w(a) = p/ρ = −1 − (a/3)·d ln ρ / da
```

```
By suitable choice of shape function S(y):

ρ(a) exhibits a broad quasi-plateau over a finite range of ln(a)

In this regime: d ln ρ / da ≈ 0 ■ w(a) ≈ −1

■ observationally indistinguishable accelerating phase.
```

B.3 Resolution of the Observational Tension

Role	Parameter	Function
Structural	$\Lambda < 0$	Global confining term; ensures bounded cycles and turnaround
Observational	Λ_eff > 0	Emergent effective parameter from ρ(a) quasi-plateau

Observers fitting late-time expansion data would infer an effective positive Λ, even though the fundamental Λ in the Hamiltonian constraint is negative. No contradiction arises, and no fine-tuning is required. This preserves the kinematical integrity of Pendulum Cosmology while remaining compatible with the empirical phenomenology of late-time acceleration.

Summary — Addendum B:

This addendum does not introduce new observational claims. It clarifies how apparent acceleration arises naturally within the existing self-dual framework while maintaining $\Lambda < 0$ as a structural necessity.

ADDENDUM

Quantum Bridge

Boundary conditions · tunneling · transfer operator · τ-lattice dynamics

This addendum strengthens the quantum/minisuperspace portion of Pendulum Cosmology by (i) making the boundary condition choice explicit as a self-adjoint extension problem, (ii) stating whether inter-cycle tunneling is forbidden or allowed as a controllable parameter, and (iii) integrating the Wheeler–DeWitt sector into the τ-lattice dynamics via a discrete transfer operator at turnaround walls.

C.1 The Quantum Problem

v4 defines $x = \ln(a)$ and treats a single cycle as a compact minisuperspace interval $x \in [-x_{max}, x_{max}]$ with a WDW-type constraint operator:

```
WDW CONSTRAINT

■ψ(x) = 0, ■ = −(■²/2M)·d²/dx² + U(x)


Missing bridge: what is the physical meaning of the boundary at ±x_max,

and how does it connect to the wall-pairing rule |U(t)| = |L(t)|?
```

C.2 Boundary Conditions as Self-Adjoint Extensions

On a finite interval, the operator $-d^2/dx^2$ is not automatically self-adjoint; it requires a boundary condition choice. The Robin family provides a natural 1-parameter family:

```
ROBIN BOUNDARY CONDITIONS

ψ′(±x_max) = η_± · ψ(±x_max) (η_± real)


Special cases:

η → ∞: Dirichlet (hard wall — the v4 choice)

η = 0: Neumann

General η: encodes how 'hard' the wall is quantum-mechanically.
```

```
Dirichlet is presented as the κ=0 / hard-wall limit, not as a fundamental
axiom.
```

C.3 Tunneling Between Cycles: Superselection vs Small Leakage

The tunneling question is answered by turning it into a model parameter:

Option A — Superselection (No Tunneling):

- Enforce decoupling by choosing BC that kills probability current at the boundary.
- Dirichlet/Neumann/Robin with real η all enforce zero net flux.
- Cycles are quantum-isolated; lattice coupling is purely classical/kinematical.

Option B — Allowed, Suppressed Tunneling:

- Treat the wall as a narrow barrier; allow small transmission amplitude.
- WKB estimate: T ~ exp(–2∫_barrier |p_x(x)| dx / ■)
- In the hard-wall limit, tunneling vanishes exponentially/identically.
- Tunneling is controllable, not mysterious.

C.4 WDW in the τ-Lattice: Quantum Transfer Rule at Walls

Promote the wavefunction to a τ-indexed family $\psi_\tau(x)$ and define a wall-localised transfer operator:

```
QUANTUM TRANSFER RULE

ψ_τ(x) defined on x ∈ [−x_max, x_max] for each τ ∈ ■


Hilbert space: ■ = ⊕_{τ∈■} L²([−x_max, x_max], dx)


Transfer operator:

■_total = ⊕_τ ■_τ + κ · Σ_τ (|τ■■τ+1| ⊗ B■ + h.c.)


κ ≥ 0: quantum coupling strength

κ = 0 recovers isolated-cycle toe-dip

B■: boundary operator (narrow support near ±x_max)

matching classical ϕ(a■,a■) wall-localisation
```

```
The pairing bijection π is folded in by letting B■ act on

the internal two-state label ε ∈ {+1,−1}:

ε(τ+1) = −ε(τ) under transfer.
```

C.5 Classical Limit: Elastic Bounce as $\kappa \to 0$

The classical rule 'elastic turnaround: ■ → −■ at the wall' becomes the semiclassical statement: in the $\kappa \to 0$ hard-wall limit, probability current reflects at $\pm x_{max}$ with phase shift determined by η. The classical bounce is recovered without adding new forces. If κ is small but nonzero, one obtains a controlled correction: rare inter-cycle tunneling events that are exponentially suppressed (WKB) and can be declared negligible for v4.

C.6 Claims by Version

Version	Quantum sector claim	Lattice coupling
v4 (current)	Real Robin BC (incl. Dirichlet); discrete spectrum; each cycle self-adjoint	$\kappa=0$: purely classical/kinematical
v5 (upgrade path)	$\kappa\neq0$ with boundary transfer operator B■ tied to ϕ's wall-localisation	Lattice transfer map implementing π on ε

Summary — Addendum C:

This addendum removes three predictable reviewer objections by: (1) explicitly parameterizing BC choices via the Robin family, (2) making tunneling a controlled on/off parameter (κ), and (3) embedding WDW into the τ-lattice via a wall transfer rule. This does not assert a completed theory of quantum gravity.

PENDULUM COSMOLOGY v4

A Cyclic Multiverse with Rules

Rux Tux · February 2026 · v4 (CPT-orientation reframe)
Purely classical & kinematical · No observational claims made

v4 reframes the R/B sector labels as discrete CPT-orientation parity states ($\varepsilon \in \{+1,-1\}$) assigned by the pairing bijection π at each turnaround wall. What propagates across the wall is orientation, not baryon number. Global CPT is conserved by the lattice sum. Baryon asymmetry is an emergent local interpretation of orientation, not a fundamental variable. The phrase 'inherited bookkeeping label' has been replaced throughout with 'orientation inherited via the pairing bijection π.' A physicist can now trace exactly what is inherited, from where, and by what mechanism.